

RESEARCH ARTICLE

Human Brain Waves Study Using EEG and Deep Learning for Emotion Recognition

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ABSTRACT Emotion Recognition is a critical area of research including healthcare, human-computer interaction, and psychology. While traditional methods mainly rely on facial expressions and textual analysis, they also have inherent flaws and cannot be reliable. Facial expression-based emotion recognition assumes that it represents genuine internal emotions that may be inaccurate. Similarly, textual analysis depends on the available data and needs help accurately capturing subtle emotions in text. However, electroencephalography (EEG) has emerged as a rising alternative for objective and real-time emotion recognition. Unlike facial and textual methods, EEG directly measures brain activity and provides a reliable result. To address this researchers have used basic machine learning methods that need manual feature extraction, which might miss essential data and make the process slow and less accurate. In this study, we propose a comprehensive methodology for EEG-based emotion recognition that addresses the limitations of traditional methods and basic machine learning techniques. Our approach involves preprocessing EEG signals using a butter-worth bandpass filter to eliminate noise, followed by feature extraction techniques. We then employ Principal Component Analysis (PCA) for dimensionality reduction, ensuring efficient data representation. To further enhance the model performance we explore machine learning classifiers (GaussianNB, SVM, Random Forest) and proposed an EEG-LSTM and GRU model with an accuracy of 97% and 96% respectively, that gives better results than the basic machine learning models.

INDEX TERMS EEG, machine learning, deep learning, FFT, neural network, PSD, spectral entropy.

I. INTRODUCTION

Affective computing has evolved as a significant subject of study to develop systems that can perceive emotions automatically. Emotion recognition technology is expected to be worth \$21.6 billion in 2019 and \$56 billion by 2024 [1]. Our daily interactions with others, our decisions, and the way we describe the world are all influenced by our emotions. Although, basic human emotion can be recognized through body language and facial expressions, but there are additional specific types of emotions [2]. In recent decades, research on emotion recognition has been robust. It enables machines to learn about human emotions, and it can be achieved primarily in two ways:

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A. NON-PHYSIOLOGICAL SIGNALS

These signals represent how our body physically operates. It contains information that we gather from our surroundings, including visual and aural data. Examples include body language, voice tone, and facial emotion.

B. PHYSIOLOGICAL SIGNALS

These signals are also related to the physical functioning of the body. It includes the changes in a human's mental state, heart rate, breathing, body temperature, and muscle tension. It can also provide us with some of the physical states, such as whether we are relaxed or stressed, and can also be used to measure the effects of different stimuli on our body.

But in actual situations, it can be very challenging to tell someone's feelings only by glancing at their face and

behavioral patterns [3]. Because, emotions are strong feelings acquired from human's happenings, moods, or relationships with others. Emotions are basically mental states that come from changes in our body and brain. They can be related to our thoughts, behaviors, or how good or bad we feel. It is difficult to recognize the real or correct emotions of every human and also the expressions of disabled people based on their appearance. Also, there is an issue in modern times, especially with teenagers and adults that they often pretend to be happy and show off (fakeness) in social settings like- parties, clubs, and gatherings with friends. During the period of the COVID-19 epidemic, this pattern has become more noticeable. Social media has a big role in this trend, as people try to present themselves as a perfect and happy life to their friends and followers, even if it is not true. They may share pictures and stories that are not genuine, which leads to a culture of comparison and competition to show off. With these virtual social gatherings on the rise, people may feel more pressure to present a happy image, even if they are struggling. In this pandemic era, there has been a growing concern about mental health and well-being, particularly among individuals who are experiencing social isolation, financial stress, and uncertainty. According to recent studies [4], the COVID-19 pandemic has led to an increase in suicidal ideation and behavior, particularly among young adults.

Emotion recognition based on EEG has the potential to be utilized as a tool to identify persons who are at risk of committing suicide. It can be achieved by identifying different patterns of brain activity that are related to depressing feelings, such as anxiety, hopelessness, and sorrow [5].

An alternate approach to analyzing how individuals feel and gaining access to human emotion can be obtained by studying brain signals [6]. There are various methods for determining their dimensions, including invasive and non-invasive approaches. The non-invasive way provides a better way of measuring brain signals because they are more convenient and comfortable for the person being measured. EEG is an example of a non-invasive method that can be used to collect those signals. It is a better way to measure brain signals compared to other techniques or methods like- fMRI and MEG. Because it can show changes in the current brain's activity over a short period and is also cheaper to set up and use.

EEG is a tool that can directly measure the neural activity of humans, and it is the most active way to assess the functions and waves of neurons. Neural activity is a type of brain function activity that occurs all at once in the brain and involves a large number of neurons in order to achieve the lowest levels of electric potential that enable EEG to record electrical signals [3]. The recorded signals are amplified and analyzed to reveal the actual patterns of brain activity known as brain waves [7]. These brain waves can change from one state to another depending on the activities that they are engaged in at a particular moment [8]. For instance, a person's brain waves in a scenario in which they are unconcerned,

easygoing, and frank are starkly different from someone who is working hard at something. On this basis, five types of brain waves that are commonly studied using EEG, where each with different frequencies and associated with different states of consciousness and cognitive processes are- 'delta waves (0.5 to 4 Hz)', 'theta waves (4 to 7 Hz)', 'alpha waves (8 to 12 Hz)', 'beta waves (13 to 30 Hz)', and 'gamma waves of 30 Hz and higher' are among the most researched waveforms [9].

C. MOTIVATION

Understanding and interpreting human emotions is crucial in various fields, including psychology, neuroscience, human-computer interaction, and affective computing. Emotion recognition has the potential to revolutionize how we interact with technology, personalize user experiences, and improve mental health diagnosis and treatment. Traditional emotion recognition methods often rely on self-reporting, which can be subjective and prone to biases. Therefore, there is a growing need for objective and accurate methods to identify and classify emotions.

By analyzing EEG signals, we can understand the neural mechanisms associated with different emotional states and develop robust models for emotion recognition.

The primary motivation behind this research paper is to explore the intersection of EEG for emotion recognition-

- Improve the understanding of the neural basis of emotion, which could have important implications for the treatment of emotional disorders such as depression and anxiety.
- Explore the effectiveness of deep learning models, such as Gated Recurrent Unit (GRU) and Long Short-Term Memory (LSTM), and analyze EEG signals for emotion recognition.
- Learn to optimize the performance of DL models in classification.
- Co-relation of emotions by analyzing the learned representations in deep learning models.

D. RESEARCH CONTRIBUTION

The research contribution is to find out the effectual and potential of integrating EEG signals and deep learning techniques for accurate and objective emotion recognition.

- Design and implement deep learning models, such as gated recurrent unit (GRU) and long short-term memory (LSTM), for analyzing EEG signals. The models will be trained and evaluated on emotion recognition tasks using appropriate datasets.
- Explore the ability of deep learning models to automatically learn meaningful representations from raw EEG signals.
- Conduct comparative analyses with traditional machine learning approaches and assess the performance gain achieved by utilizing deep learning models.

II. RELATED WORKS

Zhen et al. [10] in their study discussed the importance of face recognition and the existing literature on the neural mechanisms involved. It was necessary to do research on human participants utilizing fMRI, which recorded neural activity as the subjects performed facial recognition tasks. The authors analyzed the data using network analysis techniques to identify the hierarchical organization of brain regions involved in face recognition. The results showed that face recognition involved a hierarchical network of brain regions, with lower-level regions responsible for basic visual processing and higher-level regions responsible for more complex cognitive processes such as facial identity recognition. They discussed the importance of a hierarchical organization in the brain network for efficient face recognition. The findings offered implications for understanding the neural mechanisms of face recognition and developing more effective techniques for artificial face recognition. It did not investigate the effects of other elements, such as mood or context, on the neural network responsible for face recognition in the brain.

Pandey and Seeja [11] in their study aimed to recognize emotional states using electroencephalogram (EEG) signals through a subject-independent approach. They discussed the importance of emotion recognition in various fields and the challenges associated with traditional methods of emotion recognition. It involved collecting EEG signals from human subjects while they were exposed to emotionally evocative stimuli. They were able to classify the different emotional states by using machine learning models after training them with strategies for extracting characteristics from the EEG signals. The approach that was suggested was successful to the extent of 86.4%, in identifying six different emotional states, namely happy, sorrow, anger, fear, disgust, and surprise. This study used a limited number of emotional states, and further research may be necessary to extend the method to more complex emotional states.

Doma and Pirouz [12] explored how EEG and peripheral physiological inputs from human participants are employed in emotion recognition with ML algorithms when they were exposed to emotionally expressive stimuli. They extracted key elements from the signals using feature extraction and trained machine learning models to classify the emotional states. They employed the random forest technique, which outperforms other machine learning approaches in emotion recognition when using both EEG and periphery physiological inputs.

Tan et al. [13] in their study proposed a multimodal emotion detection system that improved the accuracy of emotion recognition carried out by integrating facial expression analysis and EEG. This study involved 20 participants who were shown emotional images while recording their facial expressions and EEG signals. They analyzed the EEG signals using a wavelet transform, then they utilized a CNN to extract information from the facial expressions. They were input into

an SVM classifier in order to make a prediction regarding the participant's feelings. The results showed that this approach achieved a higher accuracy (86.5%) in recognizing emotions than the unimodal approach that only used facial expressions (76.7%) or EEG (71.3%) alone. After combining facial expression analysis with EEG signals, this method improved the accuracy of emotion recognition. They also suggested that this method could be used in the future for the diagnosis and treatment of emotional disorders.

Hassounieh et al. [8] discussed previous studies on emotion recognition, including various methods such as physiological signals, facial expressions, and speech analysis. They also highlighted the importance of real-time emotion recognition systems in various fields, such as healthcare and HCI. They explained the feature extraction and selection process before training the ML and deep neural network approaches. The performance evaluation of the developed system using various metrics such as accuracy, sensitivity, and specificity. They also reported the high accuracy and effectiveness of the developed system in recognizing different emotional states.

Gao et al. [14] in their study introduced Deep Learning (DL) algorithms as a promising approach to overcome these challenges. Then, they explained the feature extraction process before training the DL models. The performance evaluation of the developed system using various metrics such as accuracy and confusion matrix. They highlighted the potential of DL algorithms in improving emotion recognition from EEG signals. They also suggested future research directions for improving the system's performance, including the use of more advanced deep learning architectures and the integration of multiple modalities of data.

In their study, Schyns et al. [15] explored the neural mechanisms that allowed humans to categorize facial expressions. When participants saw images of faces expressing various emotions, their brain activity was monitored using fMRI. They found that the brain processed facial expressions in a hierarchical manner, with lower-level visual features (such as eye and mouth movements) being processed first, followed by higher-level features (such as the overall expression). The brain uses a dynamic integration process to categorize facial expressions, with different stages of processing occurring at different times. They also suggested that this dynamic process may be important for the perception of complex stimuli in general, not just facial expressions.

Similarly, Ismail et al. [6] looked over the feasibility of using EEG to detect human emotions. They conducted an experiment in which participants were asked to watch short video clips designed to evoke different emotions while their brain activity was recorded using EEG. They discovered that many emotional states, including happiness, sorrow, anger, and fear, were related to patterns of brain activity in the parietal and frontal brain regions. EEG has the potential to be an effective tool for detecting human emotions. The fact that there were only 12 participants in the sample size and that it is

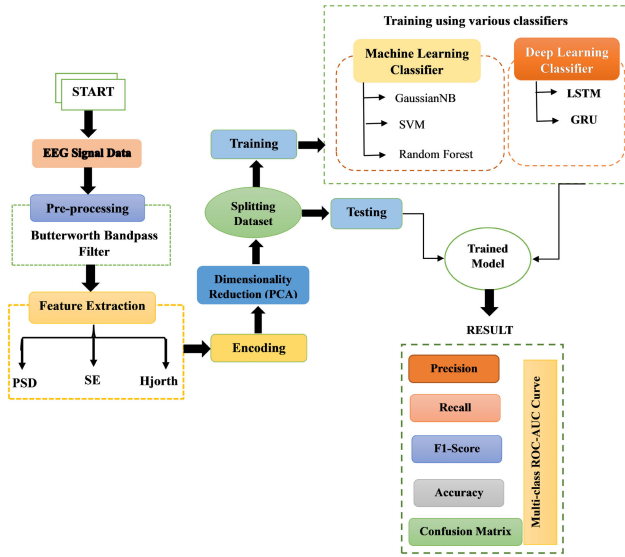


FIGURE 1. Framework for Emotion Recognition using EEG dataset.

unclear whether the findings apply to other emotional states are two limitations of this study.

III. PROPOSED METHODOLOGY

There are a few phases involved in classifying the various emotions based on the “Human Brain Waves Study using EEG and Deep Learning for Emotion Recognition” that are represented in the figure 1. The implementation involves a couple of machine learning and deep learning approaches to recognize people's real and exact emotions. The first stage is to gather the EEG data, which has been done through Kaggle¹ that contains electroencephalogram (EEG) measurements from multiple participants as they went through different states of emotions. The dataset used in this model, named the “Emotion EEG” dataset, comprises EEG recordings of the brain activity as individuals watched emotional videos. These recordings are labeled with basic emotional states: happiness, sadness, anger, fear, surprise, and neutral. Each column in the dataset contains numerical values representing brainwave amplitudes at specific time intervals. Also, the dataset contains 15 EEG channels, each capturing brainwaves from distinct locations on the scalp where electrodes were positioned.

This framework aims to predict the emotional states of individuals based on their brain activity captured through EEG measurements using the steps shown in the figure 1. The data is preprocessed to prepare it for model training. The handling of missing values, normalization of the feature values, and division of the data into training and testing sets are all included in this. The EEG signals are raw time-series data, which need to be transformed into meaningful features

for classification. Popular techniques like Fast Fourier Transform (FFT) can be used in Power Spectral Density (PSD) and Spectral Entropy (SE) to extract frequency or time-domain features from the EEG data. After feature extraction, the features are encoded into a normalized data format. Next, we used PCA which helps in minimizing the complexity of the dataset by transforming the original features into a smaller set of uncorrelated principal components. It utilizes machine learning [16] and a deep learning algorithm to train a classification model using the extracted features. The chosen model (e.g., GaussianNB, SVM, Random Forest) [17] and the deep learning models- LSTM and GRU. It is instantiated and trained using the training data. The trained model's ability to predict emotions is next tested using the testing data. To assess the model's efficiency, evaluation metrics such as “accuracy, precision, recall, and F1-score” can be generated.

EEG-LSTM Algorithm

Step 1: Loading EEG data and their corresponding emotion labels.

Step 2: Pre-processing

- i. A Butterworth bandpass filter is applied with a passband of 0.5Hz to 30Hz using-

$$\mathbf{H}(s) = \frac{1}{\sqrt{1 + ((\frac{s}{wc})^{2n})}} \quad (1)$$

where, $\mathbf{H}(s)$ = transfer function of the filter
 s =laplace variable, wc = cut-off frequency [0.5-30Hz], n =filter order (5)

- ii. Here, the signal is divided into overlapping segments using Power Spectral Density (PSD), with each segment's periodogram being calculated after applying a window function to it.

$$\mathbf{Pxx}(f) = \left(\frac{1}{M * fs} \right) |\text{FFT}(\mathbf{x}(t) * \mathbf{w}(t))|^2 \quad (2)$$

where, $\mathbf{Pxx}(f)$ = PSD estimate at frequency f ,
 M = number of segments (8), fs = sampling rate, $\mathbf{x}(t)$ = signal, $\mathbf{w}(t)$ = window function: Hamming window
 FFT denotes the discrete Fourier transform

Step 3: Feature Extraction

- i. Measurement of power of signal over a range of frequency using FFT-

$$\text{PSD} = \text{abs} \frac{(\text{fft}(\mathbf{x}(n)))^2}{N} \quad (3)$$

N = length of the signal

- ii. Spectral Entropy (SE) is calculated to measure the spectral complexity of a signal by

¹Retrieved from <https://www.kaggle.com/datasets/birdy654/eeg-brainwave-dataset-feeling-emotions?select=emotions.csv> accessed on 03/11/2022.

using PSD.

$$SE = - \sum_{m=1}^N P(m) \log_2 P(m) \quad (4)$$

where, P = normalized PSD, $P(m)$ = probability distribution of signal

- iii. To check the activity and complexity of the signal, we are using Hjorth parameter-

$$\text{Activity} = \text{var}(x(n)) \quad (5)$$

$$\text{Complexity} = \frac{\text{std}(\int \int x(n))}{\text{std}(x(n))} \quad (6)$$

var = variance, std = standard deviation of signal

Step 4: Model Creation using LSTM

- i. Initialize hidden states h_{t-1} and cell state C_t .
- ii. Compute the input, forget, and output gates, respectively-

$$\text{input gate : } i_t = \sigma(w_i [h_{t-1}, x_t] + b_i) \quad (7)$$

It regulates how information from new sources enters the cell.

$$\text{forget gate : } f_t = \sigma(w_f [h_{t-1}, x_t] + b_f) \quad (8)$$

It regulates the information transfer from the prior hidden state h_{t-1} into the present hidden state h_t

$$\text{output gate : } o_t = \sigma(w_o [h_{t-1}, x_t] + b_o) \quad (9)$$

It manages the information transfer from the cell state C_t where, “ w_i, b_i, w_f , and b_f are learnable parameters”, and “ σ is the sigmoid activation function”

- iii. Now, updating the cell state (C_t) by adding new input information and removing the old information controlled by the forget gate f_t

$$C_t = f_t * C_{t-1} + i_t * \tanh(w_c [h_{t-1}, x_t] + b_c) \quad (10)$$

where, w_c and b_c are learnable parameters, and The hyperbolic tangent activation function is $\tanh$ The final output vector is:

$$y_t = o_t * \tanh(C_t) \quad (11)$$

The cell state is using a hyperbolic tangent function and the output gate uses a sigmoid function.

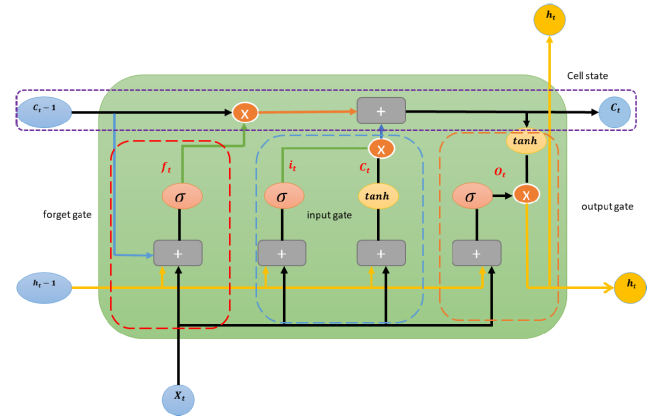


FIGURE 2. LSTM model architecture.

signal into overlapping segments, applied a window function of length 256 samples to each segment, and compute the periodogram of each segment to obtain the Power Spectral Density (PSD) estimates.

After preprocessing the feature extraction takes place, including the PSD, Spectral Entropy (SE), and Hjorth parameters (Activity and Complexity). The absolute square of the FFT of the signal, normalized by the length of the signal, is used to calculate the PSD [18]. The Shannon entropy formula is used to take the normalized PSD and calculate SE, a measure of the signal's spectral complexity [19].

The signal's variance and standard deviation, which represent the activity and complexity of the signal, respectively, are the Hjorth parameters [20]. After that, we created an EEG-LSTM and GRU model. The choice of Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) models for this study is driven by their exceptional ability to handle sequential data and capture temporal dependencies, which are crucial for effective emotion recognition from EEG signals. As illustrated in our workflow, EEG signals are inherently sequential and represent dynamic brain activity over time. LSTM and GRU networks are specifically designed to process such time-series data, making them well-suited for capturing the intricate temporal patterns associated with different emotional states. LSTM is efficient at handling 'sequential' input. LSTM networks are frequently used for tasks including speech recognition, language modeling, and time series prediction because they have the capacity to learn long-term dependencies in the input sequences. The LSTM network generates an output vector y_t and an updated hidden state vector h_t at each time step t by taking an input vector x_t and a hidden state vector h_{t-1} as inputs. The hidden state vector h_t contains information about the context of the input sequence up to time t . It has three gates - input, forget, and output gates.

Information that goes into and out of the LSTM cell in the figure 2 is managed through the help of these gates. The 'input gate' controls 'the flow of new information into the cell', 'forget gate' controls 'the transition from one

The EEG data is loaded here with their corresponding emotion labels such as: neutral, positive, and negative. A Butterworth bandpass filter with a passband of 0.5Hz to 30Hz is used to pre-process the data, and then divide the

hidden state to another', and the 'output gate' controls 'the flow of information out of the cell'. With the addition of new input data and the removal of outdated data under the direction of the forget gate, the LSTM 'cell state' is updated. By 'multiplying the cell state' by the 3 inputs, the LSTM's output is calculated (positive, negative, and neutral emotions) output gate, which is passed through a hyperbolic tangent function to obtain the final output vector [21].

Another type of recurrent neural network (RNN) designed for dealing with sequential information such as time series or data with natural language, is the GRU (Gated Recurrent Unit) in figure 3.

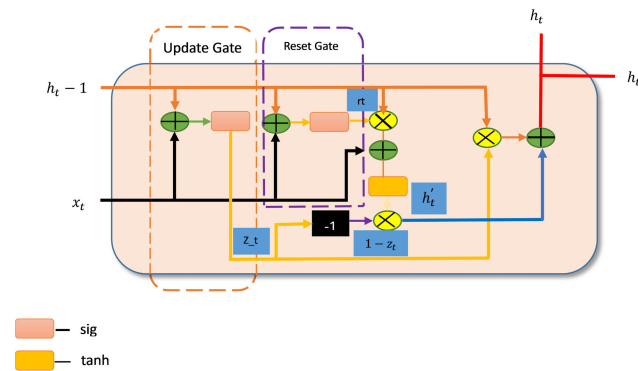


FIGURE 3. GRU model architecture.

It has been designed to solve the classic RNN's vanishing gradient issue and identify long-term dependencies in the input sequence. The 'update gate' and the 'reset gate' are its two main gates. How much of the prior memory should be kept and transferred to the current time step is decided by the update gate. It is calculated by linearly combining the 'current time step's' input and the hidden state from the 'previous time step' with a sigmoid activation function [22].

$$z_t = \sigma(W_z * x_t + U_z * h_{t-1}) \quad (12)$$

The reset gate chooses how much of the prior memory should be removed, and which elements of the prior memory should be taken into consideration in the present time step is controlled by this.

$$r_t = \sigma(W_r * x_t + U_r * h_{t-1}) \quad (13)$$

Backpropagation through time (BPTT) is used to train these by minimizing a loss function between the expected output sequence and the actual output sequence [23]. After training on the training set, the model is tested on the testing set, and measures such as "accuracy, precision, recall, and F1 score" are used to assess the model's effectiveness. Cross-validation is used to fine-tune the model's hyperparameters, such as the number of LSTM and GRU units, dropout rate, and epoch count.

IV. RESULTS AND DISCUSSION

The use of deep learning methodology and 'advanced neural networks' for EEG emotion detection and prediction

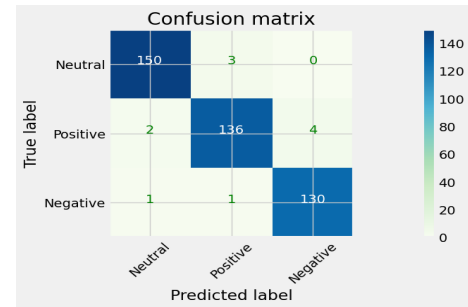


FIGURE 4. Confusion matrix of the proposed LSTM model.

produced more efficient and accurate results as in figure 4. This concept categorizes the various emotions based on EEG brain waves. The testing set's EEG signals' emotions are predicted using the trained model LSTM and GRU.

The true labels for the "Neutral" class are shown in the first row. It shows that 150 cases were correctly classified as "Neutral" (true negatives), three cases were wrongly classified as "Positive" (false positives), and 0 cases were incorrectly classified as "Negative" (false negatives). The actual labels for the "Positive" class are shown in the second row. 136 instances were properly predicted as "Positive" (true positives), 2 instances were incorrectly labeled as "Neutral" (false negatives), and 4 instances were incorrectly labeled as "Negative" (false positives). The actual labels for the "Negative" class are shown in the third row. It demonstrates that there were 130 cases where the outcome was accurately predicted as "Negative" (true negatives), 1 instance where the outcome was incorrectly calculated as "Positive," and 1 incident where both "Neutral" and "Positive" were incorrectly expected.

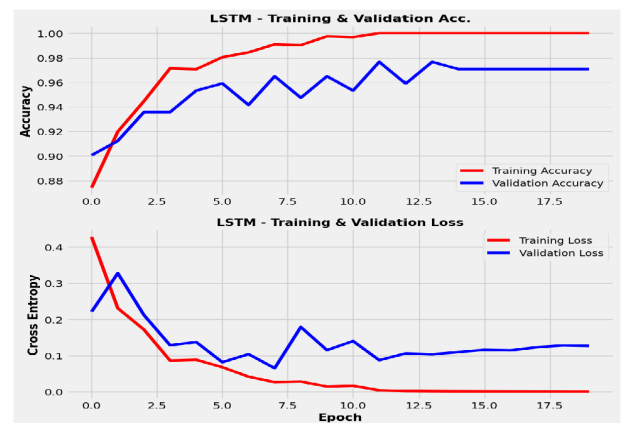


FIGURE 5. Proposed LSTM model training and validation accuracy.

In figure 5 the accuracy values are represented by the 'y-axis' on the first subplot, and the 'x-axis' is the epoch count. The blue line shows the validation accuracy, and the red line shows the training evaluation. It allows us to visualize how the accuracy of the model changes during training and validation. The second subplot's y-axis displays the loss values, which

are commonly calculated using cross-entropy, and the x-axis displays the count of epochs. The blue line is ‘validation loss’, and the red line shows ‘training loss’. It allows us to visualize the loss of the model.

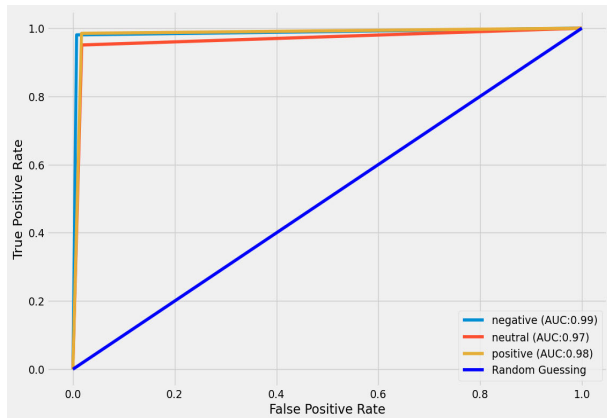


FIGURE 6. ROC curve plot for proposed LSTM model.

For each class, it illustrates the ratio between actual “positive and false positive” rates. The plot in figure 6 displays the ROC curve for each class, along with the corresponding AUC score for negative (AUC:0.99), neutral (AUC:0.97), and positive (AUC:0.98). The AUC score represents the performance of the classifier, with higher values indicating a more powerful capacity to discriminate. The legend shows the class labels and their respective AUC scores. The ‘y-axis’ represents the ‘true positive rate’, while the ‘x-axis’ represents the ‘false positive rate’.

GRU and LSTM are related since both employ different strategies for information gating to get over the vanishing gradient problem.

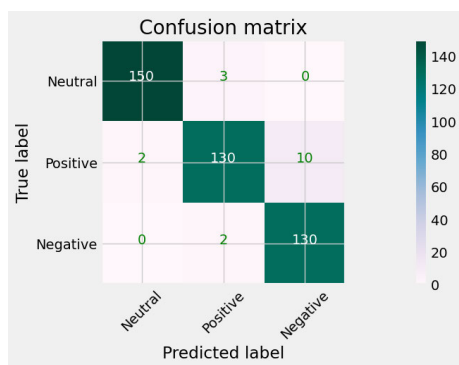


FIGURE 7. Confusion matrix of the proposed GRU model.

Figure 7 represents the evaluation of a ‘GRU classification model’. The true class is represented by the matrix's rows, while the predicted class is represented by its columns. The first row shows the 150 cases that were accurately labeled as “negative (true negatives)”, 130 instances that were accurately labeled as “neutral (true neutral)”, and 130 instances that were accurately

labeled as “positive (true positives)”. The total count of occurrences for each class that were classified “(true positives and true negatives)” and the count of instances for which the classification was incorrect (false positives and false negatives).

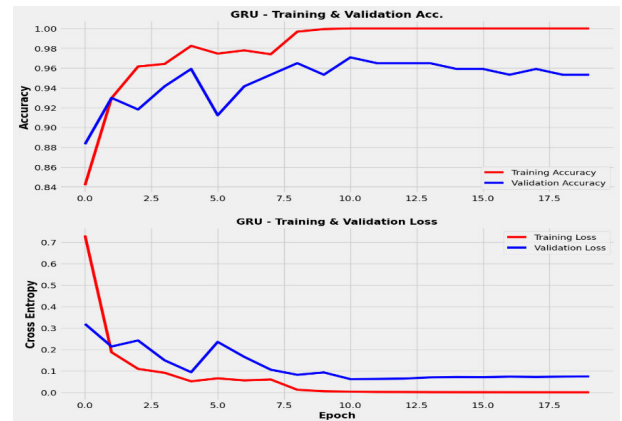


FIGURE 8. GRU model training and validation accuracy.

The “training and validation accuracy” is shown in the first subplot of figure 8 over the epochs, while the training and validation loss is shown in the second subplot. These plots provide insights into the model's performance during training, such as the convergence of accuracy and the behavior of the loss function.

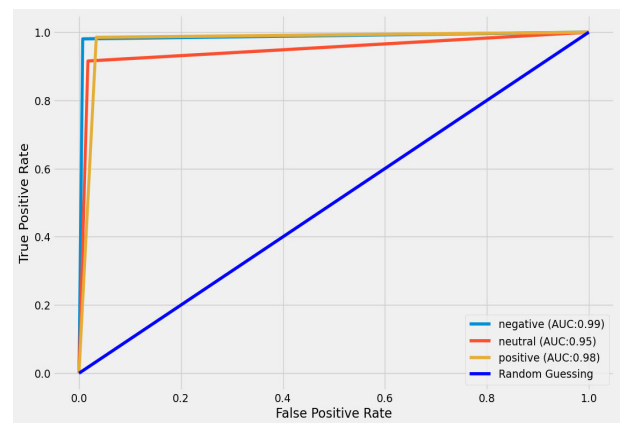


FIGURE 9. ROC curve plot for GRU model.

Each class is represented by a separate curve in figure 9, and the ‘area under curve (AUC)’ represents the classification performance for that class. The legend shows the class names and their corresponding AUC scores for negative, neutral and positive. The graph helps in determining the model's proficiency for classifying entities and evaluating performance amongst them. A higher AUC score indicates better classification accuracy.

Figure 10 represents the classification performance of the models. Each value in the matrix indicates the number of samples falling into specific categories. The diagonal

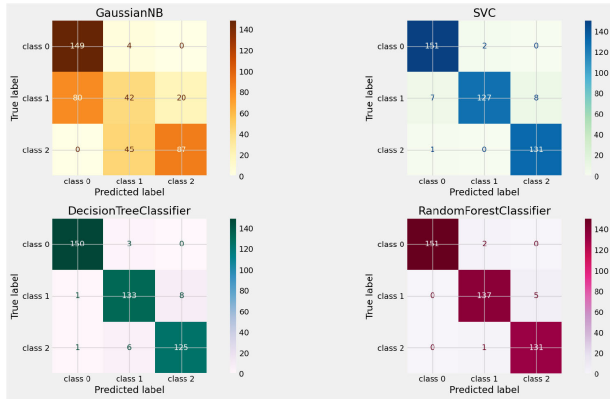


FIGURE 10. Confusion matrix of the used machine learning classifiers.

elements indicate correct predictions for each class, while off-diagonal elements represent misclassifications. Visualizing the matrices helps assess the models' performance, with different colors indicating the varying counts or percentages. By analyzing the confusion matrices, strengths and weaknesses of classifiers can be identified, aiding in selecting the most suitable model and understanding its ability to correctly classify instances.

The model's performance has been evaluated using several criteria, including- "accuracy, precision, recall, and F1 score". In Table 1, we present the performance metrics for various models.

TABLE 1. Performance model using several metrics.

Models	Accuracy	Precision	Recall	F1-score
LSTM	97	99	98	98
GRU	96	99	98	98
GaussianNB	65	65	97	78
SVM	96	95	99	97
Random Forest	96	99	98	99

The findings suggest that the LSTM model performed well in terms of accuracy and other metrics, showing that it was successful in classifying various emotions from the EEG signals. Now it shows how to classify EEG signals into various emotions using LSTM algorithms. This work has potential applications in affective computing, human-computer interaction, and neuroscience.

V. CONCLUSION AND FUTURE WORK

The model presented in this work has shown good results in classifying EEG brain signals into different emotions. By this work, the ability to classify emotions with EEG signals could enable more personalized and adaptive human-computer interaction and improve our understanding of the neural mechanisms underlying emotional processing. However, there are some limitations. The dataset was small and only included EEG data from people watching emotional videos. Additionally, the features extracted in this study are relatively

simple and may only partially capture the complex patterns in EEG signals relevant to emotion classification.

Future research could overcome these constraints by utilizing larger and more expanded datasets and exploring better feature extraction techniques and machine learning algorithms. This work represents the right step in the direction of EEG-based emotion classification systems and opens up new avenues for research in affective computing and neuroscience.

Declarations: The authors declare that they have no conflict of interest.

Data Availability Statement: The dataset analyzed during the current study is openly available at [<https://www.kaggle.com/datasets/birdy654/eeg-brainwave-dataset-feeling-emotions?select=emotions.csv>] accessed on 03/11/2022

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